KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
A Skilled and Ethical Society

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SENIOR SCHOOL CURRICULUM DESIGN

GRADE 10

GENERAL SCIENCE

2025

First Published in 2025

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FOREWORD

The Government of Kenya (GoK) is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Kenya Constitution 2010, the Kenya Vision 2030, National Curriculum Policy 2019, the United Nations Sustainable Development Goals (SDGs) and the Regional and Global conventions to which Kenya is a signatory.

Towards achieving the mission of Basic Education, the Ministry of Education (MoE) embarked on a reform process that culminated in the adoption of a Competency Based Education (CBE) system in 2017. This came after extensive research, stakeholder engagement international benchmarking and consideration of the 21st Century socio-economic realities. CBE in Kenya adopted a 2-6-3-3 structure where the numbers indicate the years learners take at various levels in Basic Education (Pre-Primary, Primary, Junior School and Senior School; respectively). CBE has for critical elements: The Competency Based Curriculum (CBC), Pedagogy (teaching and learning), Resources and the Competency Based Assessment. The MoE oversees the provision of resources while the other components (Curriculum, Pedagogy and Assessment are handled by the Kenya Institute of Curriculum Development, (KICD), Teachers Service Commission, (TSC) and the Kenya National Examinations Council (KNEC); respectively.

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KICD has successfully developed curriculum designs that have been used in the implementation of the Competency Based Curriculum (CBC) for the Pre-Primary, Primary and Juniors School levels. The first cohort of learners pursuing the CBC will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level are expected to facilitate learners to acquire relevant knowledge, skills, values and attitudes (competencies) in preparation for tertiary education and the world of work. I urge all agencies of the GoK and other stakeholders in Education to use the designs to plan for effective and efficient implementation of Competency Based Education.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Education (CBE) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The Kenya Institute of Curriculum Development (KICD) developed Competency Based Curriculum (CBC) designs that were used for pilot studies in 2017 and 2018 before the roll-out at Pre-Primary and Primary Early Year (Grades 1 to 3 in 2019. The curriculum was then introduced in the subsequent grades through *a phase in –phase out* strategy, to ensure a smooth transition. Currently, the first cohort of CBC learners are at Grade 9 – the last grade at Junior School. The implementation of CBE has since benefited immensely from the recommendations of the *Presidential Working Party on Education Reforms* (PWPER) that were released in August, 2023 and fully implemented, in terms of curriculum designs.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways, tracks and subjects of their choices guided by the potentials demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and training and/or initial competencies for the Industry. This is an important step in the realisation of the Vision and Mission of the curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They also present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

The Ministry of Education, through the State Department for Basic Education commits to provide resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop curricula, programmes and curriculum support materials for Basic and Tertiary Education and Training. The curriculum development process involves research, international benchmarking and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD,2017), that responds to the demands of the 21st Century, the United Nations Sustainable Development Goals (SDGs). African Agenda 2063, East African Community Protocol and the aspirations captured in the Kenya Constitution 2010 and the Kenya Vision 2030.

KICD receives its funding from the GoK to enable the successful achievement of the stipulated mandate and implementation of the Government and Education sector plans. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank and GoK through the *Kenya Primary Education Equity in Learning Programme* (KPEELP); a project coordinated by the MoE. Therefore, the Institute is very grateful to the GoK, through the MoE and the development partners (especially The World Bank) for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also wish to acknowledge the KICD curriculum development officers and other staff, all teachers, educators who took part as panellists; the institutions that participated and representatives of various stakeholders for their roles in the development of the Grade 10 curriculum designs. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council Chairperson and other members of the Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide the implementation of the CBE at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. **Foster nationalism and patriotism and promote national unity.**
   Kenya's people belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, which enable them to live together in harmony and foster patriotism in order to make a positive contribution to the life of the nation.

2. **Promote the social, economic, technological and industrial needs for national development.**
   Education should prepare the youth of the country to play an effective and productive role in the life of the nation.
   a) **Social Needs**
   Education in Kenya must prepare children for changes in attitudes and relationships which are necessary for the smooth progress of a rapidly developing modern economy. There is bound to be a silent social revolution in the wake of rapid modernisation. Education should assist our youth to adapt to this change.
   b) **Economic Needs**
   Education in Kenya should produce citizens with the skills, knowledge, expertise and personal qualities that are required to support a growing economy. Kenya is building a modern and independent economy which is in need of an adequate and relevant domestic workforce.
   c) **Technological and Industrial Needs**
   Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only be part of this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. **Promote individual development and self-fulfilment**
   Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

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4. **Promote sound moral and religious values.**
   Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. **Promote social equity and responsibility.**
   Education should promote social equality and foster a sense of social responsibility within an education system which provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. **Promote respect for and development of Kenya's rich and varied cultures.**
   Education should instil in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development in order to build a stable and modern society.

7. **Promote international consciousness and foster positive attitudes towards other nations.**
   Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. **Promote positive attitudes towards good health and environmental protection.**
   Education should inculcate in young people the value of good health in order for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners shall proceed to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-tertiary/university/ pre-career experience in which the learners have an opportunity to choose pathways, tracks and subjects where they shall have demonstrated interest and/or potential at the earlier levels. Senior School comprises three years of education for learners generally in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be ***engaged, empowered and ethical citizens*** ready to participate in the socio-economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise **four compulsory learning areas** namely: English, Kiswahili, Essential Mathematics/Core Mathematics and Community Service Learning (CSL). Learners will choose between Core Mathematics (for learners pursuing the Science Technology Engineering and Mathematics Pathway) and Essential Mathematics (for learners pursuing the non-STEM pathways). The learner will select three learning areas depending on their career choice, aptitude, interest and personality with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:
1. Communicate effectively and utilise information and communication technology across varied contexts.
2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems.
4. Exploit individual talents for leisure, self-fulfilment, career growth, further education and training.
5. Uphold national, moral and religious values and apply them in day-to-day life.
6. Apply and promote health care strategies in day-to-day life.
7. Protect, preserve and improve the environment for sustainability.
8. Demonstrate active local and global citizenship for harmonious co-existence.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	**Arts & Sports Science**	**Social Sciences**	**Science, Technology, Engineering & Mathematics (STEM)**
1. English 2. Kiswahili/KSL 3. Core Mathematics/ Essential Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Building Construction 32. Electricity 33. Metalwork 34. Power Mechanics 35. Wood Technology 36. Media Technology 37. Marine and Fisheries Technology

Note:
- **a)** Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.
- **b)** Every school shall offer Pastoral/Religious Programme of Instruction (P/RPI) to enhance moral, spiritual and character development of the learners.
- **c)** There shall be One lesson for personal or group study to deepen knowledge and enhance skills in areas of choice by the learners.

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Table 2 below provides guidelines on distribution of lessons at Senior School.

LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	**No of Lessons Per Week (40 minutes per lesson)**
Core Subjects 4	
1. English	5
2. Kiswahili	5
3. Essential Mathematics/Core Mathematics	5
4. Community Service Learning	3
Elective Subjects	
5. Elective 1	5
6. Elective 2	5
7. Elective 3	5
* *Physical Education*	*3*
* *ICT Skills*	*2*
**PPI*	*1*
* *Learner Personal /Group Study*	*1*
Total Number of Lessons	**40**

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ESSENCE STATEMENT

General Science focuses on biological, physical and chemical phenomena of life. The learning area seeks to equip learners with functional investigative and application skills relating to life science. It also deals with society's need to understand how the physical environment works in order to benefit from it and responsibly care for it. It is intended to develop competencies in the learner to solve societal and technological problems while taking care of the environment. The learning area utilises scientific and technological knowledge, which builds on the foundational concepts acquired from Integrated Science at Junior School.

Some of the skills relevant to the study of General Science are; classifying, communicating, measuring, designing an investigation, drawing and evaluating conclusions. The learner will need to formulate models, observe, hypothesise, identify and control variables, interpret data and make inferences based on scientific evidence. This aims at developing problem-solving and reflective skills.

These skills can support learners in navigating day-to-day life issues, crucial for success in any field of study, employment or world of work. Consequently, studying General Science provides an opportunity for the learner to develop interest in specialist fields such as sports, applied science courses and talent-based learning areas. Knowledge, skills and attitude in General Science enables the learner to develop a sense of citizenship as well as environmental management and socio-economic development. It plays an increasingly important role in the lives of all learners owing to the influence on scientific and technological development, which are necessary for the country's economic growth and the social wellbeing of its people.

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GENERAL LEARNING OUTCOMES

By the end of Senior School, the learner should be able to:

a) Acquire functional investigative and application skills relating to life science to enhance the learner's appreciation of the environment.

b) Select and use appropriate instruments to carry out basic science process skills.
c) Apply basic research and scientific skills to manipulate the environment and solve human problems.
d) Develop critical thinking skills through basic science processes and research in addressing pertinent and contemporary issues in the society.
e) Apply scientific and technological skills to enhance innovations and entrepreneurship for sustainable development.
f) Use competencies and values to promote local and global citizenship for harmonious coexistence and appreciation of biodiversity.
g) Acquire knowledge, skills and attitudes to enhance exploitation of individual talents for leisure, self -fulfilment, career growth and for further education and training.
h) Appreciate the use of effective communication and utilisation of information in scientific advancement.

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SUMMARY OF STRANDS AND SUB STRANDS

Strand	Sub Strand
1.0 Life Science	1.1 Introduction to General Science
	1.2 The Cell
	1.3 Nutrition in Animals
	1.4 Transport in Plants
	1.5 Respiration
	1.6 Plant Growth and Development
	1.7 Microorganisms
2.0 Matter and Chemical Reactions	2.1 The Periodic Table
	2.2 Chemical Families
	2.3 Chemical Bonding
	2.4 Acids, Bases and Salts
	2.5 Rates of Reactions
3.0 Natural Physical Science	3.1 Turning Effect of Force
	3.2 Linear Motion
	3.3 Waves
	3.4 Magnetism and Electromagnetic Induction

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STRAND 1.0: LIFE SCIENCE

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

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1.0 Life Science	**1.1 Introduction to General Science** - *Importance of General Science,* - *Careers in General Science* - *The principle of inference as used in science education*	By the end of the Sub Strand, the learner should be able to: a) explain the meaning of General Science as a learning area, b) outline the importance of General Science in life, environment and technology, c) identify career opportunities related to General Science, d) analyse the principle of inference in science education, e) appreciate the importance of General Science in day-to-day life.	The learner is guided to: - brainstorm on the meaning of General Science from a resource person or the internet, - use available digital and/or print media to research on scientific processes, - discuss with peers the importance of General Science in relation to human life, environment and technology, - use available digital and print media to research on career opportunities related to General Science, - draw a chart on careers related to General Science	How is General Science useful in daily life?

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			and identify with a particular career path, - explore the process of collecting evidence for drawing inferences in science education *(methods of collecting and analysing data).*	
Core competencies to be developed: - Self-efficacy: as the learner designs and presents charts on career opportunities related to General Science and identifies a particular career path of own interest. - Digital literacy: as the learner uses digital media to research on the scientific processes, importance of General Science and careers related to General Science. - Citizenship: as the learner draws a chart on careers related to General Science and identifies with a particular career path for individual and national development.				
Values: - Respect: as the learner appreciates others' opinions when discussing the importance of General Science in day-to-day life. - Responsibility: as the learner uses digital devices appropriately to search for information on the career opportunities related to General Science.				
Pertinent and Contemporary Issues (PCIs): Socio-Economic and Environmental Issues: as the learner discusses with peers the importance of General Science to the environment and technology.				

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Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**
1.0 Life Science	**1.2 The Cell** - *Types of microscopes* - *Differences between light and electron microscopes.* - *Cell structure and functions* - *Levels of cell organisation*	By the end of the Sub Strand, the learner should be able to: a) explain the differences between a light and an electron microscope. b) describe the plant and animal cell under an electron microscope, c) compare plant and animal cell structures under light and electron microscope,	The learner is guided to: - discuss with peers the differences between light and electron microscopes, - use print or non-print media to search for the structures of plant and animal cells under electron microscope, - observe photomicrographs of plant and animal cells under electron microscope to identify the components of the cells,	What makes up an organism?

		d) explain the functions of the components of plant and animal cells, e) construct a concept map of the levels of cell organisation, f) appreciate the function of the cell as a basic unit of life.	- discuss with peers the functions of components of the animal and plant cells, - draw and label the structures of animal and plant cells as seen under the electron microscope, - make a chart showing levels of cell organisation *(organelles, cells, tissues,*	

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			organs, organ systems and organisms). **Project** Use locally available materials to make a model of a plant and/or an animal cell.	

Core competencies to be developed:

- Creativity and Imagination: as the learner constructs a model of plant and animal cells as seen under the electron microscope.
- Digital literacy: as the learner searches for information from the internet on the structure of a cell as seen under the electron microscope.
- Learning to learn: as the learner reflects on the project work and adjusts appropriately.

Values:

- Respect: as the learner considers others' opinion while working on practical activities in groups such as construction of models.
- Social justice: as the learner seeks equity, equality and gender consideration in distribution of learning resources such as sharing of materials for making cell models.

Pertinent and Contemporary Issues (PCIs)

- Environmental conservation; as the learner manages wastes responsibly to avoid pollution while preparing plant and animal cell models.
- Safety and security; as the learner observes safety precautions when preparing the models.

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Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**
1.0 Life Science	**1.3 Nutrition in Animals** - *Chemical and mechanical digestion of foods* - *Adaptations of parts of the human digestive system,* - *Food tests*	By the end of the Sub Strand, the learner should be able to: a) describe the digestion of different types of food in each region of the digestive system, b) explain the adaptations of parts of the human digestive system, c) perform experiments to determine presence of different nutrients in food, d) appreciate the role played by different parts of the digestive system.	The learner is guided to: - search for information on digestion of different types of food in each region of the digestive system, - watch animation or simulation of digestion of food along the alimentary canal, - discuss the adaptations of parts of the human digestive system to their functions, - search for information from print materials and/or retrieve and save information from non-print sources on the enzymes used to break down each type of food at different regions of the digestive system and share with peers using digital presentation modes, - carry out experiments on food tests and share the results in the plenary.	How is the human digestive system adapted to its functions?

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			Project Make a model of the human digestive system using locally available materials.	

Core competencies to be developed:

- Digital literacy: as the learner retrieves and saves information from non-print sources on the enzymes used to break down each type of food at different regions of the digestive system and shares the information with peers using digital presentation modes.
- Creativity and Imagination: as the learner uses locally available materials to make a model of the human digestive system.

Values:

- Respect: as the learner considers others' opinions while discussing the adaptations of the human digestive system.
- Unity: as the learner works harmoniously when carrying out experiments on food tests.

Pertinent and Contemporary Issues (PCIs):

Safety and security: as the learner observes safety precautions as they carry out experiments on food tests.

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Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**
1.0 Life Science	**1.4 Transport in plants** - *Absorption of Water and Mineral Salts* - *Translocation* - *Factors affecting transpiration* - *Importance of transpiration*	By the end of the Sub Strand, the learner should be able to: a) describe absorption of water and mineral salt in plants, b) explain translocation in plants, c) investigate factors affecting transpiration in plants, d) describe the significance of transpiration in sustainability of plant life, e) appreciate the role of transport in plants.	The learner is guided to: - use digital devices or print media to search for information on absorption of water and mineral salts in plants, - observe plant specimen to identify structures responsible for absorption of water in plants, - use digital devices to demonstrate the adhesive and cohesive forces involved in transport in plants, - discuss with peers the process of translocation in plants, - carry out experiment(s) to investigate translocation in plants, - carry out experiments in groups to investigate factors affecting transpiration in plants *(structural aspect of the plant and environmental factors),* - sensitise the community on the importance of watering and manuring plants to sustain plant growth,	Why is transport in plants important?

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			- create awareness in the community on the use of fertilisers (*Di-ammonium Phosphate, Calcium Ammonium Nitrate, Urea*) in boosting crop production.	

Core competencies to be developed:

- Citizenship: as the learner sensitises the community on the importance of watering and manuring plants to sustain plant growth.
- Learning to learn: as the learner reflects on experiments on factors affecting transpiration in plants.

Values:

- Unity: as the learner engages in assigned roles while working with others in the process of investigation of factors affecting transpiration.
- Patriotism: as the learner sensitises the community on the importance of watering and manuring plants to sustain plant growth,

Pertinent and Contemporary Issues (PCIs)

- Safety and security: as the learner observes safety precautions while carrying out experiments in groups to investigate factors affecting transpiration in plants.
- Environmental issues: as the learner creates awareness in the community on the use of fertilisers (Di-ammonium Phosphate, Calcium Ammonium Nitrate, Urea) in boosting crop production.

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Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**
1.0 Life Science	**1.5 Respiration** - *Aerobic and anaerobic respiration* - *Respiratory substrates and respiratory quotient* - *Factors affecting respiration* - *Economic importance of anaerobic respiration*	By the end of the Sub Strand, the learner should be able to: a) explain the meaning of respiration in living things, b) describe aerobic and anaerobic respiration in living things, c) relate the respiratory quotient to the type of substrate and type of respiration in living things, d) explain factors affecting respiration in living things, e) describe economic importance of anaerobic respiration at home and in industry, f) appreciate the significance of respiration in living things.	The learner is guided to: - discuss with peers the meaning of respiration in living things, - discuss with peers the difference between aerobic and anaerobic respiration in living things and present in plenary, - use print or non-print media to search for information on effects of anaerobic respiration in living things, - carry out activities to investigate anaerobic respiration in living organisms *(fermentation using yeast cells).* - discuss and test the properties of fermented products (acidity), - calculate the Respiratory Quotient to determine the type of substrate and type of respiration in living things, - discuss with peers the factors affecting respiration in living things,	How is aerobic and anaerobic respiration important in a society's economy?

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			- search for information on the economic importance of anaerobic respiration in living things. **Project** Carry out a project that utilises the concept of anaerobic respiration such as making of yoghurt, porridge, biogas. **CAUTION!** *No brewing of alcoholic substances.*	

Core competencies to be developed:

- Communication and collaboration: as the learner discusses with peers the factors affecting energy requirements in human beings.
- Learning to learn: as the learner carries out activities to investigate anaerobic respiration in living organisms.

Values:

- Integrity: as the learner observes ethical behaviour when working on a project of value addition of food products that utilises the concept of aerobic respiration.
- Peace: as the learner works harmoniously with others in carrying out a project that utilises the concept of anaerobic respiration such as making of yoghurt, porridge, biogas.

Pertinent and Contemporary Issues (PCIs)

- Financial literacy: as the learner undertakes a project that utilises the concept of anaerobic respiration in value addition of food products such as milk and flour
- Environmental issues: as the learner develops the skill of making biogas as an alternative source of fuel (energy).

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Life Science	**1.6 Plant growth and development** - *Concept of growth and development in plants* - *Seed dormancy* - *Conditions necessary for germination* - *Types of germination* - *Primary and secondary growth in plants* - *Role of growth hormones in plants*	By the end of the Sub Strand, the learner should be able to: a) explain the meaning of growth and development in plants, b) describe the causes of seed dormancy in plants, c) investigate the conditions necessary for germination, d) differentiate between epigeal and hypogeal germination, e) distinguish between primary and secondary growth in plants, f) describe factors that contribute to growth and development in plants, g) describe the role of growth hormones in plants,	The learner is guided to: - use digital or print media to search for differences between growth and development in plants, - discuss with peers causes of seed dormancy in plants, - search for information on how to break seed dormancy in plants and share with peers, - carry out experiments with peers to investigate conditions necessary for germination, - carry out experiments to investigate epigeal and hypogeal germination, - discuss with peers the differences between primary and secondary growth in plants, - discuss with peers the role of water, temperature, light and nutrients in the growth and development in plants,	Why is the concept of growth in plants important?

		h) appreciate the concept of growth and development in plants.	- use print and non-print media to search for information on the role of growth hormones in plants.	

Core competencies to be developed:

- Communication and collaboration: as the learner shares information gathered on differences between plant growth and development.
- Self-efficacy: as the learner works on practical activities to investigate epigeal and hypogeal germination.
- Learning to learn: as the learner reflects on the experiments investigating conditions necessary for germination.

Values:

Responsibility: Learner engages in assigned roles while working with others in the process of learning such as investigating conditions necessary for germination.

Pertinent and Contemporary Issues (PCIs)

Cyber security: as the learner observes online safety when searching for information on growth and development of plants.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Life Science	**1.7 Microorganisms** - *Types of microorganisms* - *Mode of transmission and infections of different microorganisms* - *Prevention and control of microorganisms*	By the end of the Sub Strand, the learner should be able to: a) identify microorganisms that affect human beings, b) explain modes of transmission of microorganisms in human beings, c) identify types of infections caused by microorganisms in human beings, d) describe methods of control and prevention of infections caused by microorganisms, e) explain the economic importance of microorganisms,	The learner is guided to: - discuss types of microorganisms that affect human beings *(fungi, bacteria and viruses),* - use digital or print media to search for information on the modes of transmission of microorganisms in human beings, - use digital or print media to research on infections caused by microorganisms in human beings *(fungal, bacterial and viral infections),* - discuss how the mode of transmission of microorganisms determine their control and prevention measures/methods, - discuss the economic importance of microorganisms with peers. **Project** 1. Investigate growth of microorganisms such as formation of moulds on food substrates.	How are micro-organisms important in day-to-day life?
		f) appreciate the effect of microorganisms in day-to-day life.	2. Sensitise the community on the significance of food preservation.	

Core competencies to be developed:

- Digital literacy: as the learner uses digital devices to research on infections caused by microorganisms as well as their control and prevention mechanisms.
- Communication and collaboration: as the learner shares information gathered on types of microorganism infections, control and prevention.

Values:

Respect is promoted as the learner becomes considerate of others' opinion while discussing methods of controlling and preventing microorganism infections.

Pertinent and Contemporary Issues (PCIs)

Socio-economic issues (poverty eradication); as the learner sensitises the community on the significance of food preservation and the economic importance of microorganisms.

Suggested Assessment Rubric

Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
Ability to outline the importance of General Science in human life, environment and technology	Correctly outlines the importance of General Science in human life, environment and technology, giving some examples from the locality	Correctly outlines the importance of General Science in human life, environment and technology	Correctly outlines some of the importance of General Science in human life, environment or technology	Can hardly outline the importance of General Science in human life, environment or technology
Ability to analyse the principle of inference in science education	Comprehensively and correctly analyses the principle of inference in science education	Correctly analyses the principle of inference in science education	Correctly analyses the principle of inference in science education less comprehensively	Can hardly analyse the principle of inference in science education
Ability to describe the plant and animal cell under an electron microscope	Comprehensively and correctly describes the plant and animal cell under an electron microscope	Correctly describes the plant and animal cell under an electron microscope	Correctly describes the plant or animal cell under an electron microscope less comprehensively	Can hardly describe the plant or animal cell under an electron microscope
Ability to relate the respiratory quotient to the type of substrate and respiration in living things	Comprehensively and accurately relates the respiratory quotient to the type of substrate and respiration in living things	Accurately relates the respiratory quotient to the type of substrate and respiration in living things	Accurately relates the respiratory quotient to the type of substrate and respiration in living things less comprehensively	Can hardly relate the respiratory quotient to the type of substrate and respiration in living things

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Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
Ability to investigate factors affecting transpiration	Comprehensively investigates all factors affecting transpiration	Investigates all factors affecting transpiration	Investigates most of the factors affecting transpiration	Investigates a few of the factors affecting transpiration
Ability to determine the conditions necessary for germination	Comprehensively and accurately determines the conditions necessary for germination	Accurately determines the conditions necessary for germination	Accurately determines most of the conditions necessary for germination	Determines few conditions necessary for germination.
Ability to perform experiment on test for reducing sugars as per the following procedure: 1. Put about 2 cm^3 of test solution in a test tube 2. Add an equal amount of Benedict's solution and mix 3. Place the test tube with the mixture in a hot water bath for a few minutes	Systematically and correctly performs experiment on test for reducing sugars following all the steps in the procedure	Correctly performs experiment on test for reducing sugars following all the steps in the procedure	Performs experiment on test for reducing sugars following any three of the steps in the procedure	Performs experiment on test for reducing sugars following any two of the steps in the procedure

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Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
Observe and record all the colour changes				
Ability to explain modes of transmission of microorganisms in human beings	Comprehensively and accurately explains modes of transmission of microorganisms in human beings	Accurately explains modes of transmission of microorganisms in human beings	Accurately explains most of the modes of transmission of microorganisms in human beings	Explains a few of the modes of transmission of microorganisms in human beings

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STRAND 2.0: MATTER AND CHEMICAL REACTIONS

Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**
2.0 Matter and Chemical Reactions	**2.1 The Periodic Table** *Classification of the first twenty elements of the periodic table* • *Electron configuration of atoms of elements* • *Ion formation* • *Valency and oxidation number* • *Chemical formulae of common compounds* • *Chemical equations*	By the end of the Sub Strand, the learner should be able to: a) examine electron arrangement of an atom of an element, b) classify elements into groups and periods of the periodic table, c) explain the stability and electron affinity of atoms of elements, d) describe ion formation for the first 20 elements of the periodic table, e) formulate chemical formula of common compounds, f) write balanced equations of chemical reactions, g) appreciate the importance of the structure of the atom in the development of the periodic table.	The learner is guided to: • carry out activities to review the concepts of the structure of an atom, atomic number, mass number and the electron arrangement of atoms of selected elements, • arrange the first 20 elements using similarities in their electron arrangements, • discuss with peers the order adopted in developing the periodic table *(the periodic law),* • discuss with peers how atoms of elements acquire stability *(losing or gaining electrons)* relating to the atom's ability to attain the positive or negative charge *(cations and anions),* • discuss the unreactive nature of the noble gases, • draw dot (.) and cross (x) diagrams to illustrate ion formation for the first 20 elements of the periodic table, • discuss with peers on how to determine the valency and oxidation numbers of elements and radicals (*ammonium ion, sulphate, nitrate, hydroxide, carbonate, phosphate, hydrogen carbonate*), • write chemical formula of common compounds using valencies of elements and radicals, • use valencies of elements and radicals to balance chemical equations, • watch animations and simulations on ion formation and atomicity of gases using digital devices. **Project** 1. Model electron configuration of any five elements of the periodic table (1 - 20) using locally available materials. 2. Make a chart of common substances that are made from the elements and radicals (*include names and formulae*)	1. Why is electron arrangement significant? 2. How are elements arranged in the periodic table?

Core competencies to be developed:

- Critical Thinking and Problem Solving: as the learner sorts and arranges the first twenty elements of the periodic table according to their similarities.
- Creativity and Imagination: as the learner models the electron configuration of any five elements of the periodic table using locally available materials.

Values:

- Respect: as the learner recognises and respects the input of every group member as they discuss how to determine the valence and oxidation numbers of elements and radicals.
- Unity: as the learner takes turns in playing roles when drawing dot (.) and cross (x) diagrams to illustrate ion formation for atoms of elements of the periodic table.
- Responsibility: as the learner uses digital devices appropriately to watch simulations and animations on ion formation.

Pertinent and Contemporary Issues (PCIs)

- Life skills (Problem solving skills): as the learner arranges the first twenty elements of the periodic table into groups and periods.
- Financial literacy: as the learner uses locally available materials as a way of minimising the expenses when modelling the electron configuration of any five elements of the periodic table.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Matter and Chemical Reactions	**2.2 Chemical Families** • *Alkali metals, alkaline earth metals, halogens, noble gases and transition metals* • *Physical and chemical properties of selected elements* • *Uses of the elements and their compounds*	By the end of the Sub Strand, the learner should be able to: a) identify chemical families in the periodic table, b) classify elements of the periodic table into their chemical families, c) investigate the physical and chemical properties of selected elements of the periodic table, d) outline the uses of elements of the periodic table, e) appreciate the uses of the elements of the periodic table in day-to-day life.	The learner is guided to: • search for information on the categories of elements in the periodic table *(alkali, alkaline earth metals, halogens, noble gases and transition metals),* • sort and group the first twenty elements of the periodic table into their families • discuss with peers selected transition elements of the periodic table *(copper, iron, zinc, lead),* • observe safety when carrying out activities to investigate properties of elements of the periodic table. Appropriately use first aid kits in case of injuries, • carry out activities to investigate physical properties of alkali metals, alkaline earth metals, halogens, noble gases and transition metals, • carry out activities to investigate the reaction of sodium, magnesium and	How are elements suited to their use in day-to-day life?

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			chlorine with air (*oxygen*), water and dilute acids where applicable (*write balanced chemical equations for the reactions)*, • discuss with peers the properties of elements of the periodic table in relation to their uses, • discuss with peers how noble gases are suited to their applications in day-to-day life, • discuss with peers the properties of elements that make them suitable for use in illuminating roads, • use available digital and print media to research on transition elements and their uses.	
Core competencies to be developed: • Self-efficacy: as the learner carries out activities to investigate the reaction of sodium, magnesium and chlorine with air and dilute acids. • Digital literacy: as the learner uses available digital and print media to research on transition elements and their uses.				
Values: Love: as the learner takes care of self and others by observing safety when carrying out activities to investigate properties of elements of the periodic table and appropriately using first aid kits in case of injuries.				
Pertinent and Contemporary Issues (PCIs) Road safety: as the learner discusses with peers the properties of elements that make them suitable for use in illuminating roads.				

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Strand	**Sub Strand**	**Specific Learning Outcomes**	**Suggested Learning Experiences**	**Suggested Key Inquiry Question(s)**
2.0 Matter and Chemical Reactions	**2.3 Chemical bonding** • *Bond and bond types* • *Bond formation in common compounds and metals* • *Bond types and their structures* • *Uses of Diamond, Graphite and Aluminium.*	By the end of the Sub Strand, the learner should be able to: a) illustrate chemical bonding in common compounds, b) identify bond types in common compounds, c) distinguish structures of selected elements and compounds, d) describe the relationship between the bond types and the physical properties of elements and compounds, e) select appropriate materials for use based on their bond type and structure, f) reflect on the appropriateness of substances used in day-to-day life.	The learner is guided to: • discuss with peers the role of valence electrons in bond formation, • search for information from online or other available resources on the bond types of common compounds *(water, sodium chloride, hydrogen gas, oxygen gas, Iodine, Diamond and Graphite),* • navigate through online sources to access virtual laboratories for simulations on formation of chemical bonds, • discuss with peers the formation of different types of chemical bonds *(ionic, covalent, dative-covalent, hydrogen, metallic and Van der Waals),* • use dots (**.**) and crosses (**x**) to illustrate chemical bonding for selected molecules and compounds, • observe safety precautions when carrying out activities to investigate	How do bond types relate to physical properties of compounds?

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			properties of elements and compounds, • carry out activities to demonstrate physical properties of different elements and compounds (*melting point, boiling point, solubility, electrical and thermal conductivity*), • discuss with peers the relationship between bond types and physical properties of different structures *(giant ionic, simple molecular, giant atomic and giant metallic),* • explore with peers the uses of elements and compounds based on their bond types and structures *(diamond, graphite and Aluminium).* **Project** 1. Model structures of molecules and compounds using locally available resources. 2. Sensitise the community on use and care of common appliances such as utensils and other cookware (Aluminium).	

Core competencies to be developed:

- Communication and collaboration: as the learner sensitises the community on use and care of common appliances such as utensils and other cookware.
- Digital literacy: the learner navigates through online sources to access virtual laboratories for simulations on formation of chemical bonds.
- Creativity and imagination: as the learner undertakes a project of modelling ball and stick ionic and covalent structures of selected molecules using locally available resources.

Values:

- Respect: as the learner recognises and appreciates the input of every group member when discussing the relationship between bond types and physical properties of different structures.
- Patriotism: as the learner sensitises the community on use and care of common appliances such as utensils and other cookware.

Pertinent and Contemporary Issues (PCIs):

- Safety and Security: as the learner observes safety precautions when carrying out activities to investigate properties of elements and compounds.
- Financial Literacy: as the learner makes informed decisions when selecting appropriate items, appliances and equipment made from specific elements and compounds.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Matter and Chemical Reactions	**2.4 Acids, Bases and Salts** • *Acids and bases* • *Reactions of acids and bases* • *Salts*	By the end of the Sub Strand, the learner should be able to: a) Distinguish between acids and bases using the universal indicator and pH chart, b) explain the role of acids and bases in biological and chemical processes, c) identify products of chemical reactions of acids and bases, d) classify salts according to their behaviour when exposed to air, e) outline applications of salts in real-life situations,	The learner is guided to: • discuss with peers the meaning of acids and bases *(limited to the presence of hydrogen ions and hydroxide ions for acids and bases respectively),* • test acidic and basic substances in the environment using the universal indicator and pH chart to determine their pH levels *(bleaching agents, vinegar, salts, soda, juices, water, wood ash, antacids, lemons),* • discuss with peers the functions of acids and bases in biological and chemical processes *(digestion, respiration, bleaching action),* • carry out experiments and present findings on chemical properties of dilute acids and bases *(reaction of acids with bases, acids with carbonates and metals)* • carry out experiments to investigate the behaviour of salts when exposed to air *(hygroscopy, deliquescence and efflorescence),*	1. Why are salts important in day-to-day life? 2. How do salts behave when exposed to air?

		f) promote proper use of salts in day-to-day life.	• listen attentively to peers when discussing the various applications of salts *(agriculture, food industry, medicine, laundry, road use),* • discuss with peers the effect of salts on the environment *(eutrophication, soil and air pollution),* • use digital or print media to research on the applications of salts in day-to-day life. **Project** 1. Sensitise community on the acidity and basicity of substances used in day-to-day life such as bleaching agents, vinegar, salts, soda, juices, water, wood ash, antacids, lemons among others. 2. Create awareness on the importance of putting required amounts of salt on food to avoid cases of high blood pressure.	

Core competencies to be developed:
- Communication and collaboration: as the learner sensitises the community on the acidity and basicity of substances used in day-to-day life such as bleaching agents, vinegar, salts, soda, juices, water, wood ash, antacids, lemons among others.
- Self-efficacy: as the learner creates awareness on the importance of putting required amounts of salt on food to avoid cases of high blood pressure.
- Digital literacy: as the learner uses digital devices to research on the applications of salts in day-to-day life.

Values:
- Responsibility: as the learner carefully uses digital devices to research on the applications of salts in day-to-day life.
- Respect: as the learner respects each other's opinion while discussing with peers the meaning of acids, bases and salts.

Pertinent and Contemporary Issues (PCIs):
- Socio-economic and environmental education: as the learner discusses the effect of salts on the environment.
- Safety and security: as the learner carries out experiments and presents findings on chemical properties of dilute acids and bases.
- Learner support programme (peer education): as the learner discusses with peers the functions of acids and bases in biological and chemical processes *(digestion, respiration, bleaching action).*

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
2.0 Matter and Chemical Reactions	**2.5 Rates of Reactions** • *Rates of reactions* • *Factors affecting the rates of reactions*	By the end of the Sub Strand, the learner should be able to: a) describe rate of chemical reactions, b) perform experiments involving the rate of chemical reactions, c) explain factors that influence the rate of reactions in biological, chemical and physical processes, d) appreciate the importance of manipulating factors affecting the rate of chemical and biological processes in day-to-day life.	The learner is guided to: • discuss with peers the meaning of the rate of a chemical reaction, • observe safety when carrying out activities to investigate rates of reactions, • carry out experiments to investigate the rate of chemical reactions, • carry out experiments to investigate factors that affect the rate of biological, chemical and physical processes; *(temperature, pressure, catalyst, light, concentration, Surface area)*, • discuss with peers the factors that affect the rate of reactions in biological, chemical and physical processes, • use available digital devices and print media to research on the importance of optimum conditions in biological, chemical and physical processes.	How do various factors affect the rate of reactions?

Core competencies to be developed:

- Communication and collaboration: as the learner discusses with peers the factors that affect the rate of chemical reactions.
- Digital literacy: as the learner uses digital devices and print media to research on the importance of optimum conditions in biological, chemical and physical processes.

Values:

- Responsibility: as the learner carefully uses digital devices to search for information on the importance of optimum conditions in industrial processes.
- Respect: as the learner respects others' opinion while discussing with peers in a group the importance of efficient industrial processes.

Pertinent and Contemporary Issues (PCIs):

Safety and security: as the learner observes safety measures when carrying out activities to investigate rates of reactions.

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Suggested Assessment Rubric

Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
Ability to classify the first 20 elements into groups and periods of the periodic table.	Classifies all the first 20 elements into groups and periods of the periodic table.	Classifies more than 15 elements into groups and periods of the periodic table.	Classifies at least 10 elements into groups and periods of the periodic table.	Classifies less than 5 elements into groups and periods of the periodic table.
Ability to outline the uses of elements of the periodic table	Exhaustively outlines the uses of elements of the periodic table	Outlines the uses of elements of the periodic table	Outlines most uses of elements of the periodic table	Outlines the uses of a few elements of the periodic table
Ability to describe the relationship between the bond types and the physical properties of elements and compounds	Comprehensively and clearly describes the relationship between the bond types and the physical properties of elements and compounds, with illustrations	Clearly describes the relationship between the bond types and the physical properties of elements and compounds	Describes the relationship between most of the bond types and the physical properties of elements or compounds	Describes the relationship between a few of the bond types and the physical properties of elements or compounds
Ability to identify products of chemical reactions of acids and bases	Identifies all products of chemical reactions of acids and bases and helps peers	Identifies products of chemical reactions of acids and bases	Identifies most of the products of chemical reactions of acids and bases	Identifies a few of the products of chemical reactions of acids and bases.

Level Indicator	Exceeds expectations	Meets expectations	Approaches expectations	Below expectations
Ability to explain factors that influence the rate of reactions in biological, chemical and physical processes	Comprehensively and clearly explains all factors that influence the rate of reactions in biological, chemical and physical processes	Clearly explains factors that influence the rate of reactions in biological, chemical and physical processes	Explains most of the factors that influence the rate of reactions in biological, chemical or physical processes	Explains a few factors that influence the rate of reactions in biological or chemical or physical processes

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STRAND 3.0: NATURAL PHYSICAL SCIENCE

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Natural Physical Science	**3.1 Turning effect of force** • *Moments of a force at a point* • *Moments of anti-parallel forces* • *Applications of the turning effect of force in real life.*	By the end of the Sub Strand, the learner should be able to: a) determine moments of a force at a point, b) calculate the moments of anti-parallel forces, c) apply the turning effect of force in real-life situations, d) appreciate the importance of the turning effect of force in everyday life.	The learner is guided to: • use digital devices and/or print media to research on moments of a force at a point, • watch animations and simulations on moments of anti-parallel forces and discuss with peers, • demonstrate the principle of moments of force by balancing a metre rule, • solve numerical problems involving both moments of a force and moments of anti-parallel forces, • walk around the school to identify application(s) of turning effect of force, • watch video clips on how the concept of turning effect of force is applied in real-life situations, • discuss with peers other instances where the turning effect of force can be utilised.	How is the turning effect of force used in our daily life?

Core competencies to be developed:

- Critical thinking and problem solving: as the learner solves numerical problems involving both moments of a force and moments of anti-parallel forces and identifies areas where moments of a force is used in day-to-day life.
- Digital literacy: as the learner interacts with digital technology when researching on moments of a force at a point and moments of anti-parallel forces.

Values:

- Integrity: the learner displays honesty while using digital devices to search for information on moments of a force at a point and moments of anti-parallel forces.
- Respect: the learner respects others' opinions while discussing with peers on the examples involving turning effect of force in daily life.

Pertinent and Contemporary Issues (PCIs):

Life skills (self – awareness): as the learner demonstrates the principle of moments of force by balancing a metre rule.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)

3.0 Natural Physical Science	**3.2 Linear motion** - *Terms used in linear motion* - *Equations of linear motion (no derivations needed),* - *Examples of linear motion with constant acceleration* - *Freely falling bodies.*	By the end of the Sub Strand, the learner should be able to: a) explain terms used in linear motion, b) calculate variables of motion using equations of linear motion with constant acceleration, c) investigate the effect of gravity on bodies under free fall, d) appreciate the significance of linear motion and free fall in real life.	The learner is guided to: - discuss with peers the meaning and differences between the terms used in linear motion *(distance, displacement, speed, velocity and acceleration),* - use digital devices/print media to research on the equations of linear motion, - solve numerical problems in linear motion with constant acceleration using equations of linear motion, - discuss with peers the applications of linear motion in real life and present in plenary, - drop objects from a certain height to observe the increase in speed of the objects and infer the effect of gravity on objects,	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day to day life be mitigated?

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			- observe safety measures when dropping objects to investigate the effect of gravity on objects, - discuss the dangers of accelerating on sloping surfaces such as steeply sloping roads, - discuss with peers the safety measures put in real-life situations to minimise fatal injuries resulting from free fall, - carry out assignments on examples of bodies under free-fall motion.	

Core competencies to be developed:
- Communication and collaboration: as the learner works with peers to discuss and present in plenary the application of linear motion in real life.
- Learning to learn: as the learner carries out assignments on examples of bodies under free-fall motion and presents the findings in a group discussion.

Values:
- Peace: the learner displays tolerance and calmness when discussing the application of linear motion in real life.
- Unity: the learner displays cooperation and collaboration with others while investigating the effect of gravity on objects.

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Pertinent and Contemporary Issues (PCIs):
- Road safety: increased awareness on how to minimise fatal injuries resulting from road crashes on sloping roads using the concept of free fall of objects.
- Life Skills Education (self-awareness): the learner observes safety measures when dropping objects to investigate the effect of gravity on objects.

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Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Natural Physical Science	**3.3 Waves** - *Terms used in waves* - *The wave equation* - *Properties of waves* - *Application of properties of waves in real life.*	By the end of the Sub Strand, the learner should be able to: a) explain the terms used in waves, b) interpret the wave equation as used in science, c) calculate the wave characteristics using the wave equation, d) demonstrate the effects of properties of waves in real life, e) appreciate the applications of waves in real life.	The learner is guided to: - use print or non-print media to research on the terms used in waves and properties of waves *(wavelength, period, amplitude, frequency and velocity)*, - discuss with peers the interpretation of the wave equation by relating the characteristics of the wave, - solve numerical problems on wave characteristics using the wave equation, - carry out experiments to investigate reflection of sound waves using the echo method, - watch video clips on refraction of sound waves both during the day and night and illustrate the concept diagrammatically, - carry out experiments to establish that sound waves can be heard round a corner, - search for information on the effect of waves on the environment *(communities, marine ecosystems, infrastructure and operations)*, - discuss with peers, the mitigation measures of the effects of waves on the environment, - search for information on the applications of waves in day-to-day life such as safe road use *(reflection, refraction and diffraction)* and discuss with peers.	1. How are waves applicable in our day-to-day life? 2. How do waves affect our environment?

Core competencies to be developed:

- Learning to learn: as the learner searches for information on applications of reflection, refraction and diffraction of waves.
- Critical thinking and problem solving: as the learner searches for information on mitigation measures of the effects of waves on the environment *(communities, ecosystems, infrastructure and operations)*

Values:

Unity: as the learners play different roles when carrying out experiments to investigate reflection of sound waves using the echo method.

Pertinent and Contemporary Issues:

- Safety and security: as the learner appreciates some of the applications of waves related to safety in transport.
- Environmental issues: as the learner searches for information on the effect of waves on the environment *(communities, marine ecosystems, infrastructure and operations)*.

Strand	Sub Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Natural Physical Science	**3.4 Magnetism and electromagnetic induction** • *Magnetisation and demagnetisation* • *Magnetic fields* • *Electromagnetic induction* • *Factors affecting the magnitude of induced electromotive force (e.m.f) (qualitative treatment),* • *Application of electromagnetic induction.*	By the end of the Sub Strand, the learner should be able to: a) describe the methods of magnetisation and demagnetisation in soft iron, b) describe the magnetic field patterns around a magnet, c) describe induced electromotive force in electromagnetism, d) perform an experiment on electromagnetic induction, e) explain factors affecting the magnitude of induced electromotive force, f) appreciate the applications of electromagnetic induction in day-to-day life.	The learner is guided to: • carry out experiments on magnetisation and demagnetisation in soft iron *(electrical, induction, hammering and stroking),* • discuss the observations made in the methods to determine the most suitable method of magnetisation, • carry out experiments on demagnetisation *(electrical, hammering and heating),* • utilise online or other available resources to explore the direction of magnetic fields, • draw the magnetic field patterns around a magnet, • discuss with peers the meaning of the term "induced electromotive force" in electromagnetism, • carry out an experiment to demonstrate electromagnetic induction using a U-shaped magnet, galvanometer and a straight conductor, • use available digital and print media to research on the factors that affect the magnitude of induced electromotive force, • discuss with peers the ways that electromagnetic induction can be applied in day-to-day life. **Project** Design and make a simple electric bell using locally available materials.	How is the study of electromagnetic induction important in our daily life?

Core competencies to be developed:
- Creativity and imagination: as the learner designs and makes a simple electric bell using locally available materials.
- Learning to learn: as the learner carries out experiments of magnetising a magnetic material and demagnetising a magnet.

Values:
- Unity: as the learner carries out assigned roles when performing experiments to demonstrate electromagnetic induction.

- Responsibility: as the learner takes care of the apparatus (U-shaped magnet, galvanometer and straight conductors) while carrying out experiments on electromagnetic induction.

Pertinent and Contemporary Issues:
- Life skill; (self-esteem): as the learner explains the methods of magnetisation and demagnetisation.

Suggested Assessment Rubric

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Level Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to apply the turning effect of force in real-life situations	Correctly applies the turning effect of force in real-life situations giving appropriate examples	Correctly applies the turning effect of force in real-life situations	Applies the turning effect of force in some real-life situations	Has challenges applying the turning effect of force in real-life situations
Ability to apply the properties of waves in real-life situations	Applies the properties of waves in real-life situations giving appropriate examples	Applies the properties of waves in real-life situations	Applies most of the properties of waves in real-life situations	Applies a few of the properties of waves in real-life situations
Ability to experiment on electromagnetic induction using the following procedure: 1. Using a sensitive galvanometer, a straight copper conductor, a U-shaped magnet and connecting wires, connect the apparatus to demonstrate electromagnetic induction, 2. Observe the deflection in the galvanometer when the conductor is: a) stationary between the poles, b) moved parallel to the direction of the magnetic field, c) moved vertically upwards between the poles, d) moved at an angle to the direction of the magnetic field.	Experiments electromagnetic induction following all the procedures stepwise and draws all the expected conclusions with further explanations	Experiments electromagnetic induction following all the procedures stepwise and draws expected conclusions	Experiments electromagnetic induction following all the procedures or draws most of the expected conclusions	Experiments electromagnetic induction following some of the procedures or draws a few expected conclusions

APPENDIX: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Assessment Methods in Science	Learning Resources	Non-Formal Activities
• Reflections • Game Playing • Pre-Post Testing • Model Making • Explorations • Experiments • Investigations • Conventions, Conferences and Debates • Teacher Observations • Project • Journals • Portfolio • Oral or Aural Question(s) • Learner's Profile • Written Tests • Anecdotal Records	• Laboratory apparatus and equipment • Textbooks • Models • Digital media (Radio and TV education programmes, Kenya Education Cloud and OERs) • Print media (charts, pictures, journals, magazines) • Digital Devices • Software • Recordings • Resource persons	• Visiting the science historical sites. • Organising walks to have live learning experiences. • Developing simple guidelines on how to identify and solve some community problems. • Conducting science document analysis. • Participating in talks by resource persons on science concepts. • Participating in science clubs and societies. • Attending and participating in Science and Engineering fairs. • Organising and participating in exchange programmes. • Making oral presentations and demonstrations on science issues.

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